



Indian Institute of Technology Jodhpur

Department of Mathematics

Mathematics II MAL1020

Assignment I

AY 25-26

Semester II

Full Marks 50

Deadline: Feb 5, 2026

1. A square matrix A is invertible if and only if A is a product of elementary matrices. (5)
2. Determine the values of $a, b, c, d \in \mathbb{R}$ such that the following system of equations has a solution:

$$x_1 + x_2 + x_3 - x_4 = a$$

$$x_1 - x_2 - x_3 + x_4 = b$$

$$x_1 + x_2 + x_3 + x_4 = c$$

$$x_1 - x_2 + x_3 + x_4 = d.$$

(5)

3. A matrix of the form

$$V = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ x_1 & x_2 & x_3 & \cdots & x_n \\ x_1^2 & x_2^2 & x_3^2 & \cdots & x_n^2 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ x_1^{n-1} & x_2^{n-1} & x_3^{n-1} & \cdots & x_n^{n-1} \end{pmatrix}$$

is known as a Vandermonde matrix. Prove that

$$\det(V) = \prod_{j=2}^n \left\{ \prod_{i=1}^{j-1} (x_j - x_i) \right\}.$$

(5)

4. Determine if $x^3 + x^2$, $x^2 + x$, $x + 1$ span $\mathcal{P}_3(\mathbb{R})$. Check whether the above vectors are linear dependence/independence property of these vectors. Do they form a basis for the above vector space? (5)
5. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be vectors in $\mathbb{R}^n$. Show that the above vectors form a basis for $\mathbb{R}^n$ if and only if the matrix $(\mathbf{v}_1 \ \mathbf{v}_2 \ \cdots \ \mathbf{v}_n)$ is nonsingular. (5)
6. Let $T : \mathbb{R}^5 \rightarrow \mathbb{R}^4$ be a linear transformation defined by

$$T(x_1, x_2, x_3, x_4, x_5) = \begin{pmatrix} x_1 + 2x_2 + x_3 + \alpha x_5 \\ 2x_1 + 4x_2 + 2x_3 + \alpha x_5 \\ x_1 + 2x_2 + x_3 + \alpha x_5 \\ x_4 \end{pmatrix},$$

where $\alpha \in \mathbb{R}$.

- (a) Determine the rank of T for all values of α .
- (b) Find the nullity of T for all values of α .
- (c) Identify the values of α for which the dimension of the null space increases.

(5)

7. Let

$$T : \mathbb{R}^6 \rightarrow \mathbb{R}^4, \quad S : \mathbb{R}^4 \rightarrow \mathbb{R}^3$$

be linear transformations satisfying the following conditions:

- (a) $\text{rank}(T) = 3$,
- (b) $\dim(\ker T) = 3$,
- (c) $\text{rank}(S) = 2$,
- (d) $\ker S \cap \text{Im } T \neq \{0\}$.

Define the composition $U = S \circ T : \mathbb{R}^6 \rightarrow \mathbb{R}^3$.

- (a) Determine all possible values of $\text{rank}(U)$.
- (b) Find $\dim(\ker U)$.
- (c) Explain why $\text{rank}(U) \neq \text{rank}(S) \text{rank}(T)$.

(5)

8. Let P_3 denote the vector space of all real polynomials of degree at most 3, and let $M_{2 \times 2}(\mathbb{R})$ denote the vector space of all 2×2 real matrices.

Define a linear transformation

$$T : P_3 \longrightarrow M_{2 \times 2}(\mathbb{R})$$

by

$$T(p(x)) = \begin{pmatrix} p(0) & p'(0) \\ p(1) & p'(1) \end{pmatrix}.$$

- (a) Prove that T is a linear transformation.
- (b) Find a basis for $\ker(T)$.
- (c) Find a basis for $\text{Im}(T)$ and hence compute the rank of T .
- (d) Is T injective? Is it surjective? Justify your answers.

(5)

9. Let $W = \mathbb{R}^5$. Consider the subspaces

$$U = \text{span} \{(1, 2, 1, 0, 1), (2, 4, 2, 0, 2), (0, 1, 1, 1, 0)\}$$

and

$$V = \text{span} \{(1, 0, -1, 1, 0), (0, 1, 0, -1, 1)\}.$$

- (a) Find a basis for U and determine $\dim(U)$.
- (b) Find a basis for V and determine $\dim(V)$.
- (c) Compute the intersection $U \cap V$.
- (d) Determine whether U and V are complementary subspaces of $\mathbb{R}^5$.
- (e) If U and V are not complementary, construct a subspace $V' \subset \mathbb{R}^5$ such that

$$\mathbb{R}^5 = U \oplus V'.$$

- (f) Express the vector

$$x = (3, 5, 2, 1, 4)$$

uniquely as $x = u + v'$, where $u \in U$ and $v' \in V'$.

(5)

10. Show that $\mathbb{R}$ is a vector space over $\mathbb{Q}$. Find the dimension of this vector space.

(5)